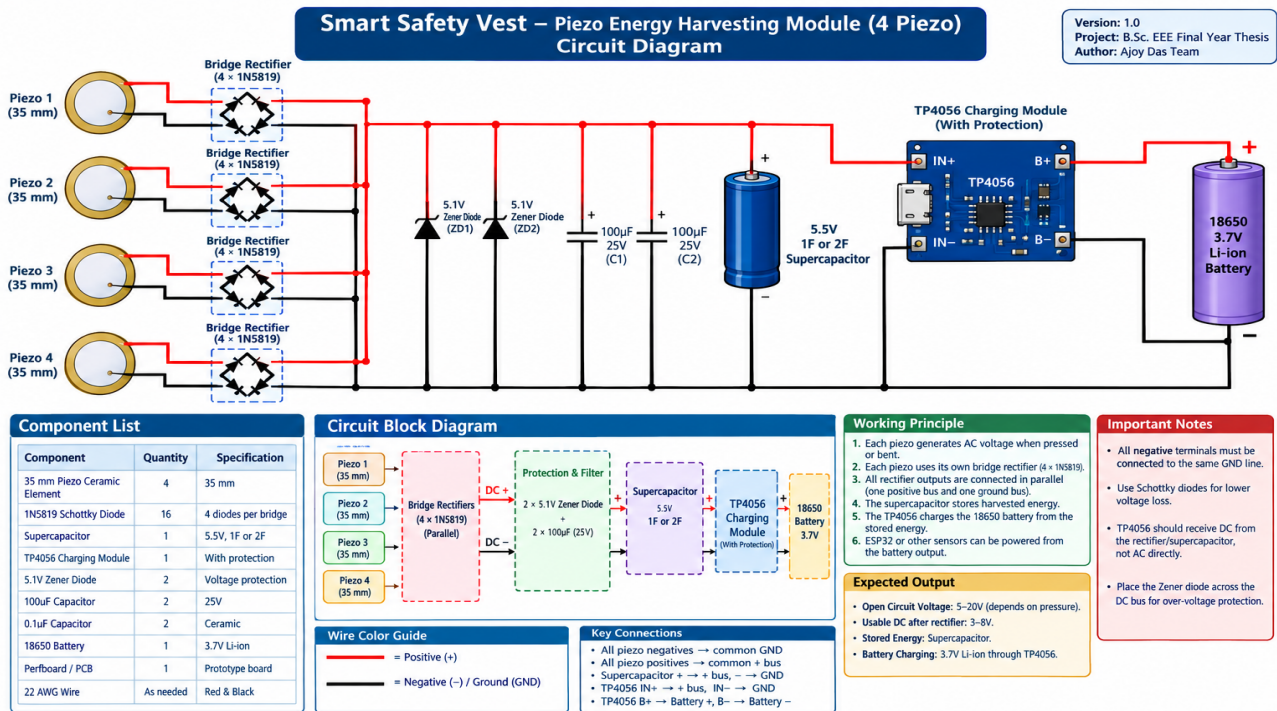


SMART SAFETY VEST

Piezo Energy Harvesting Module

Four-Piezoelectric Disc Energy Harvesting and Battery Charging Subsystem



Project Type	B.Sc. EEE Final Year Thesis
Project	Smart Safety Vest for Industrial Workers
Module	Piezo Energy Harvesting Module
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Table of Contents

1. Introduction
2. Objectives
3. Required Components
4. Circuit Configuration
5. Detailed Circuit Connections
6. Voltage Protection and Filtering
7. Supercapacitor Storage
8. TP4056 Charging Section
9. 18650 Battery Connection
10. Wiring Color Convention
11. Circuit Block Diagram
12. Working Principle
13. Expected Output
14. Safety and Connection Considerations
15. Conclusion

1. Introduction

The Smart Safety Vest project is designed to improve industrial worker safety by integrating sensing, communication, and power-management technologies into a wearable safety system.

As part of the project, a Piezo Energy Harvesting Module is proposed to generate electrical energy from mechanical pressure, vibration, or bending experienced by the vest during worker movement. The proposed module uses four 35 mm piezoelectric discs. Each piezoelectric disc generates an AC electrical voltage when mechanical force is applied. Individual bridge rectifiers convert the generated AC voltage into DC voltage, after which the rectified outputs are connected in parallel to form a common positive and negative DC bus.

The harvested electrical energy is intended to be stored in a 5.5 V supercapacitor and subsequently supplied to a TP4056 charging module for charging a 3.7 V 18650 Li-ion battery.

2. Objectives

- Harvest electrical energy from mechanical movement and pressure.
- Use four piezoelectric discs for energy generation.
- Convert the generated AC voltage into DC using bridge rectifiers.
- Combine the rectified outputs into a common DC bus.
- Store harvested energy in a supercapacitor.
- Provide the stored energy to the battery-charging section.
- Obtain a rechargeable power source for the Smart Safety Vest system.

3. Required Components

Sl.	Component	Qty.	Specification
1	Piezoelectric Disc	4	35 mm
2	1N5819 Schottky Diode	16	4 diodes per bridge
3	Supercapacitor	1	5.5 V, 1 F or 2 F
4	TP4056 Charging Module	1	With protection
5	Zener Diode	2	5.1 V
6	Electrolytic Capacitor	2	100 μ F, 25 V
7	Ceramic Capacitor	2	0.1 μ F
8	18650 Battery	1	3.7 V Li-ion
9	Perfboard/PCB	1	Prototype board
10	Connecting Wire	As required	22 AWG, Red & Black

4. Circuit Configuration

The proposed circuit follows the sequence: Piezoelectric Energy Generation → Bridge Rectification → Protection & Filtering → Supercapacitor Storage → TP4056 Charging Module → 18650 Battery.

Each of the four piezoelectric discs has its own bridge rectifier consisting of four 1N5819 Schottky diodes. The four rectifier outputs are connected in parallel so that all positive outputs form one common positive DC bus and all negative outputs form one common negative/GND bus.

5. Detailed Circuit Connections

5.1 Piezoelectric Discs

Piezo 1 → Bridge Rectifier 1; Piezo 2 → Bridge Rectifier 2; Piezo 3 → Bridge Rectifier 3; Piezo 4 → Bridge Rectifier 4.

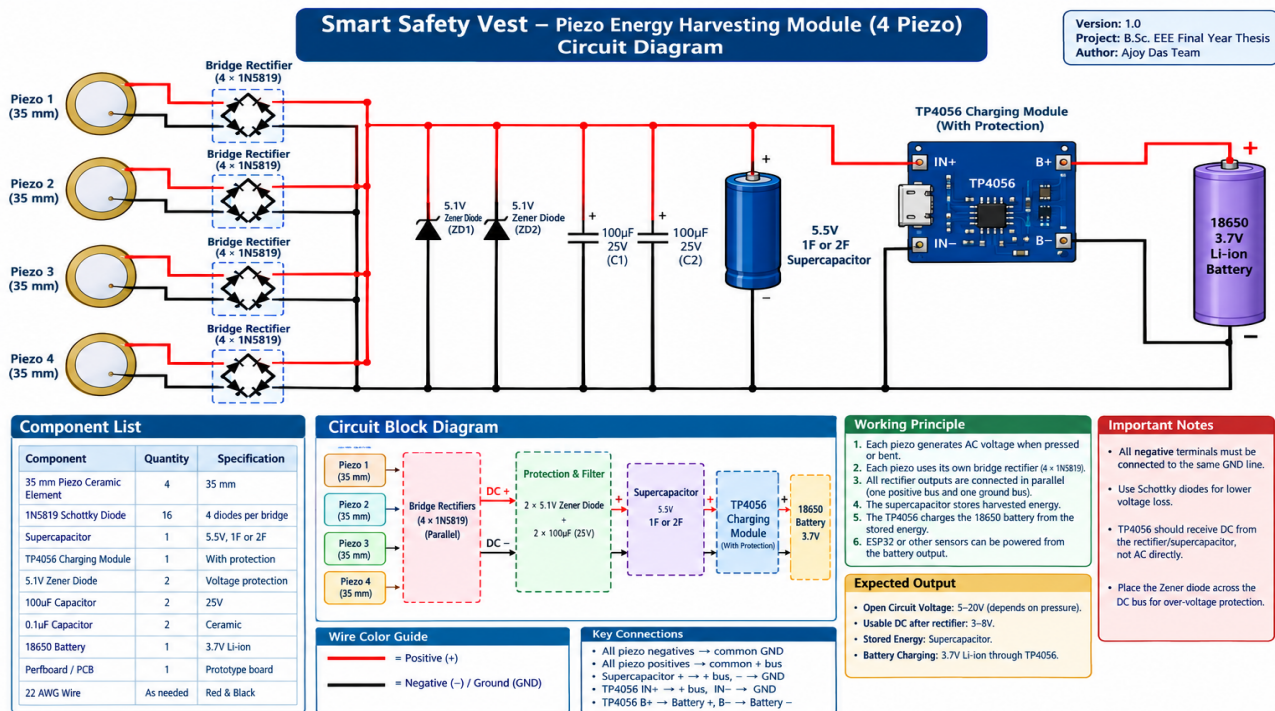
5.2 Bridge Rectifiers

Each piezoelectric disc uses a separate full-bridge rectifier made from four 1N5819 Schottky diodes. Therefore, four bridges require 16 diodes in total.

5.3 Common DC Bus

Rectifier 1 (+), Rectifier 2 (+), Rectifier 3 (+), and Rectifier 4 (+) are joined to the common DC positive (+) bus. Rectifier 1 (–), Rectifier 2 (–), Rectifier 3 (–), and Rectifier 4 (–) are joined to the common DC negative (–)/GND bus.

5.4 Circuit Diagram



6. Voltage Protection and Filtering

Two 5.1 V Zener diodes are included for voltage protection. The protection section is connected across the DC bus. Two 100 µF, 25 V capacitors and two 0.1 µF ceramic capacitors are included in the filtering section.

The purpose of this section is to provide voltage protection and filtering before the energy reaches the storage and charging stages.

7. Supercapacitor Storage

A 5.5 V, 1 F or 2 F supercapacitor is used as the energy-storage element. The supercapacitor is connected across the common DC bus: DC (+) → Supercapacitor (+), and DC (–) → Supercapacitor (–).

8. TP4056 Charging Section

The stored DC energy is supplied to the TP4056 charging module. Supercapacitor/DC Positive (+) → TP4056 IN+; Common DC Negative (–) → TP4056 IN–. The supplied project specification states that the TP4056 should receive DC from the rectifier/supercapacitor section, not raw AC directly from the piezoelectric discs.

9. 18650 Battery Connection

TP4056 B+ → Battery (+); TP4056 B– → Battery (–).

10. Wiring Color Convention

Wire Color	Connection
Red	Positive (+)
Black	Negative (-) / GND

Positive power path: Piezo Rectifier (+) → Common DC (+) → Supercapacitor (+) → TP4056 IN+.

Negative power path: Piezo Rectifier (-) → Common GND (-) → Supercapacitor (-) → TP4056 IN-.

Battery path: TP4056 B+ → Battery (+); TP4056 B- → Battery (-).

11. Circuit Block Diagram

4 × Piezo Discs → 4 × Individual Bridge Rectifiers (4 × 1N5819 per bridge) → Common DC (+)/(-) Bus → 5.1 V Zener Protection + Capacitor Filtering → 5.5 V Supercapacitor → TP4056 Charging Module → 3.7 V 18650 Li-ion Battery → Smart Safety Vest Electronics.

12. Working Principle

1. Mechanical pressure or movement is applied to the piezoelectric discs.
2. The piezoelectric discs generate AC electrical voltage.
3. Each piezoelectric disc sends its AC output to an individual bridge rectifier.
4. The bridge rectifier converts the AC voltage into DC voltage.
5. The four rectifier outputs are connected in parallel.
6. The generated DC energy passes through the protection and filtering section.
7. The electrical energy is stored in the 5.5 V supercapacitor.
8. The stored DC energy is supplied to the TP4056 charging module.
9. The TP4056 provides the charging interface for the 3.7 V 18650 battery.
10. The battery can subsequently provide power to the Smart Safety Vest electronics.

13. Expected Output

Parameter	Expected Value
Number of Piezo Elements	4
Piezo Size	35 mm
Open-Circuit Voltage	Approximately 5–20 V, depending on pressure
Usable DC after Rectification	Approximately 3–8 V
Energy Storage	5.5 V supercapacitor
Battery	3.7 V 18650 Li-ion
Charging Interface	TP4056 module

14. Safety and Connection Considerations

- All positive connections must remain on the positive DC bus.
- All negative connections must remain on the common GND bus.
- Check the polarity of the supercapacitor before connection.
- Check the polarity of electrolytic capacitors before soldering.

- Verify the anode/cathode orientation of every 1N5819 diode according to the bridge-rectifier configuration.
- Do not interchange TP4056 IN+ and IN–.
- Do not interchange TP4056 B+ and B–.
- Do not connect raw piezoelectric AC directly to the TP4056.
- Place the Zener protection section across the DC bus as specified.
- Use red wire for positive connections and black wire for negative/GND connections.

Practical validation note: this report preserves the supplied project specification. Before permanent soldering or battery charging, the actual assembled circuit should be electrically tested and the TP4056 input requirements, protection arrangement, and measured voltage/current levels should be validated on the hardware.

15. Conclusion

The proposed four-piezo energy harvesting module provides a structured method for collecting electrical energy generated from mechanical movement of the Smart Safety Vest. Individual bridge rectifiers convert the AC output of the four piezoelectric elements into DC, and the outputs are combined into a common DC bus.

The harvested energy is then processed through the protection and filtering section, stored in a 5.5 V supercapacitor, and supplied to the TP4056 charging section for the proposed 3.7 V 18650 battery subsystem. This report provides the documented circuit structure and connection plan for further prototype implementation and testing.